Drawing with Sound

Can we make a vibrating plate draw *any* picture we want?

A plain-language tour of our Chladni-plate project — written so a curious high-schooler can follow every step.

1. A 230-year-old magic trick

Sprinkle fine sand on a thin metal plate, then stroke its edge with a violin bow (or vibrate it with a speaker). At certain pitches the sand suddenly leaps around and freezes into a beautiful, symmetric pattern — stars, grids, flowers.

It feels like magic, but it is pure physics. When the plate vibrates, some lines on it barely move at all. These are the **nodal lines** (the "still lines"). Sand gets bounced off the parts that shake hard and piles up exactly on the still lines. So the pattern you see is a hidden "shape of the vibration" made visible. These are called **Chladni figures**, after Ernst Chladni, who showed them off in the 1780s.

> **Picture it:** the plate is like a trampoline. Some spots bounce a lot, some spots barely move. Sand rolls off the bouncy spots and collects on the calm ones, drawing the "map" of where the trampoline is quiet.

2. Our big, slightly crazy question

Every pitch gives a different pattern. So we asked a bold question:

Instead of accepting whatever pattern appears, can we work backwards — and ***design a plate on purpose*** *so the sand draws a picture we choose?*

We wanted the sand to spell our school initials, "IC", or draw an X, a cross, a ring. The plan: keep the plate a fixed 15 cm square, shake it from a single point in the very centre, and only change *how thick the plate is* at each spot (imagine a 15×15 grid of little tiles, each a different thickness). A computer would search for the thickness map that makes the sand draw our target. This "work backwards from the answer" idea is called **inverse design**.

3. The surprising answer: physics says "no" — and that's the interesting part

We threw everything at it: hundreds of computer simulations, about twenty different strategies, even a 36-hour no-sleep coding push. And the honest result was: **you cannot make a centrally-shaken square plate draw just any picture.** Not because we weren't clever enough — because the laws of physics forbid it.

Here is the heart of why. Our plate is a perfect square, and we shake it from the exact centre. That setup is *extremely symmetric*: you can rotate it by 90° or flip it like a mirror and it looks the same. A deep rule of physics says: **if you push a symmetric thing**

from its symmetric centre, only patterns that share that same symmetry are allowed to appear.

> **Picture it:** pushing a kid on a swing. If you always push straight from behind, you can make the swing go higher and higher — but no matter how hard you push, you will *never* make it spin sideways. The *direction* you push decides what motions are even possible. Shaking a plate dead-centre is the same: it can only "push" certain symmetric patterns into existence.

4. Why "X" is allowed but "IC" is impossible

We built a little piece of maths that measures how much of a target picture is "symmetric enough" to be allowed. The results were blunt:

- An **X shape** is **100%** allowed — it has the same four-fold symmetry as the plate, so in principle it can appear cleanly.
- A **single diagonal line** is only about **half** allowed.
- The letters **"IC"** are only about **42%** allowed. The other **58%** of that picture is physically forbidden — it simply has no way to form, no matter how good the computer is.

This was a genuinely useful discovery, even though it sounds negative. For the first time we could say *exactly* why "IC" would never show up, with a real mathematical reason rather than "our algorithm isn't good enough yet". (The grown-up name for this rule, if you want to impress a physics teacher, is *symmetry and irreducible representations*.)

5. The clever cheat that didn't work

We thought we had found a loophole. 3D-printed plastic has tiny fibres all running one way, so the plate can be slightly stronger in one direction than another (engineers call this **anisotropy**). If we rotated the "grain" differently in each tile, maybe we could break the perfect symmetry and sneak forbidden patterns in.

On paper it looked brilliant. In reality it did almost nothing. In the range of pitches we use, the plate's behaviour is dominated by its *weight* (how heavy each part is), and the fibre direction only affects its *stiffness* — a much weaker effect here. The grain's influence was drowned out completely.

> **Picture it:** trying to steer a speeding freight train by blowing on it. Your breath is a real force — but next to the train's momentum it changes nothing.

The punchline is almost funny: cheap, ordinary plastic (PLA) that any school 3D-printer can use worked *as well or better* than expensive carbon-fibre plastic. We deleted the fancy "grain" feature entirely so nobody would waste time on it again.

6. The fast calculator that liked to lie

To search through millions of possible plates quickly, we used a **simplified model** — a rough, fast sketch of the physics. To check the real answer, we used a slow but accurate professional simulator (COMSOL).

The catch: the fast sketch was an optimist. It would promise a pattern 8–18 times clearer; the accurate simulator then delivered only 1.5–4 times. The gap wasn't random noise — it came from real differences (the sketch treats the plate as paper-thin; the real one has thickness, a proper clamp, full 3D behaviour).

> **Picture it:** a fast sketch is like a friend who says "trust me, this shortcut is way faster!" — great for pointing you in a direction, terrible as the final judge. So we used the fast model as a **compass**, never as the **referee**. The accurate simulator always had the final word.

7. The long road: how we kept failing forward

Real research is messy. Ours went through three "eras":

Era 1 — blind guessing.
Just let the computer try millions of thickness maps. It only ever produced generic blobs and stars, never letters. Lesson: brute force wasn't going to cut it.

Era 2 — getting smart.
We added physics-based filters and a way for the computer to follow "which direction makes this better". Scores jumped about five-fold — but the pictures *still* didn't look like the target.

Era 3 — changing the goal.
We finally accepted "looks similar" instead of "looks identical", and re-defined success using the real sand physics. That's when we discovered we'd been grading ourselves with the *wrong ruler* for months, and that some of our best results had been hiding in plain sight.

Along the way we learned some lessons worth keeping:

- **Honesty beats pretty numbers.** We once had a flashy score of 0.2; when we measured honestly it dropped to 0.034. We kept the honest number.
- **Be willing to kill your favourite idea.** The "fancy grain" cheat was our pet theory. The data said it was useless, so it had to go.
- **Tiny bugs can hide the truth.** One almost-invisible rounding number in our code quietly squashed our real results for weeks until we found it.
- **Hit the note exactly.** Driving the plate slightly off its natural pitch let the centre push dominate and ruined the pattern; hitting the exact pitch made clean lines snap into view.

8. So what did we actually build?

Not a magic "any picture you want" machine — that doesn't exist. Instead we built something honest and genuinely useful:

1. A tool that **tells you upfront** whether the picture you want is even physically possible (its "symmetry score").
2. A two-step pipeline that pushes each *possible* target as far as the real physics allows — first letting the accurate simulator pick the best vibration, then fine-tuning the plate's thickness.
3. A finished design you can **send straight to a 3D printer** and hold in your hand.

We wrapped it all in an app with buttons, so a user can draw a target, hit "run", and see what physics will and won't give them.

9. The real lesson

We started out wanting to make sand spell anything. We ended up proving that's impossible — and that turned out to be the better prize.

*Sometimes the most valuable result of a science project isn't getting the thing you wanted. It's understanding **exactly why** the thing you wanted is impossible — and then building something real, honest, and useful right up against that limit.*

And if you ever want to push past the wall? The physics is clear about that too: stop shaking the plate from the centre. Add more shakers, or push from off to one side, or change the plate's shape. Break the symmetry on purpose — and a whole new world of patterns opens up. But that's a project for next summer.

There is a full technical paper behind this friendly version, with all the equations, data tables, and references — ask if you'd like to dive into the deep end.